



Review Article

Exercise and lipids

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ABSTRACT

Evidence from the existing literature suggests that exercise has positive effects for prevention and treatment of cardiovascular diseases by reducing risk factors such as elevated blood lipids. Based on clinical and observational clinical trials, it is well established that increased physical activity and regular exercise has a favourable impact on blood lipids and lipoprotein profiles. Exercise training significantly decreases blood triglycerides concentration and increases high density lipoprotein cholesterol levels. Though the Indian data depicting the effect of exercise on lipids is scarce, exercise directly improves “atherogenic dyslipidaemia” which is frequently present among Indians i.e. HDL-C is increased, TG is reduced and LDL-C particle size is improved. While drug therapy is key to the treatment of dyslipidaemia, lifestyle alterations such as exercise should continue to be actively promoted and encouraged by clinicians. Exercise is a low cost, non pharmacological therapeutic lifestyle change that is of value to lipid metabolism and cardiovascular fitness.

1. Introduction

Apart from drug therapy, aerobic exercise, which is easier to do and has fewer side effects, has been shown to improve the prognosis of cardiovascular disease (CVD). Aerobic exercise is defined as any form of physical activity that produces an increased heart rate and respiratory volume to meet the oxygen requirements of the activated muscle.¹ In a prospective cohort study of exercise and lipid metabolism, it was found that the risk of mortality is significantly reduced when combined with statin therapy and aerobic exercise compared to either alone.²

2. Exercise and Lipids

2.1. High density lipoprotein-cholesterol

According to the Health Risk Factors Exercise Training and Genetics (HERITAGE) family Study, which is the largest reported study on the contribution of endurance exercise training to blood lipid parameters, showed $3.6 \pm 11\%$ increase in HDL-C concentrations compared with baseline after 20 weeks of supervised exercise in 675 healthy normolipidemic subjects.³ This increase in HDL-C primarily involved the HDL2 fraction, with an associated increase in ApoA-1.

Data are inconsistent regarding the relationship between the intensity of exercise and increase in HDL-C. According to HERITAGE

Family Study, exercise training-induced plasma lipid improvements may be genotype-dependent.

In another study of 111 sedentary overweight men and women with mild-to-moderate dyslipidaemia, it was found that high amount/high intensity exercise significantly increased HDL-C by 8.8 % and HDL particle size and diameter also increased resulting in more beneficial HDL2 fraction.⁴ It was concluded that the improvements were related to the amount of exercise and not to the intensity of exercise.

2.2. Triglycerides

It appears that exercise affects serum triglyceride levels favourably especially in patients with TG-HDL-C axis abnormality such as seen in metabolic syndrome subjects with high baseline TG and low HDL-C. In a subset analysis of 200 men enrolled in HERITAGE Family Study showed that 22 weeks of exercise reduced the plasma TG levels by 15 % especially in men with low HDL, elevated TGs and abdominal obesity.⁵

2.3. Low density lipoprotein-cholesterol

Low density lipoprotein cholesterol is the most common predictor of CV events but exercise training alone has shown no significant effect of LDL-C. However, physical exercise appears to increase the average size of LDL-C particles and reduce the number of atherogenic small dense

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LDL-C particles.^{4,6} This is especially relevant for Indian population who are known to have increased fraction of small dense LDL.

If individuals are able to perform exercise for longer duration, it can lead to metabolic improvements such as shift of more atherogenic small, dense LDL fraction towards larger LDL particles in a dose-dependent manner through 3.5–7 h of moderate–vigorous exercise per week or 30–60 min of exercise daily.

2.4. Total cholesterol

Total cholesterol levels do not change unless, there is a change in dietary pattern or a reduction in body weight after exercise training.

2.5. Non-high density lipoprotein cholesterol

Current evidence suggests that non-HDL-C is a better indicator of CVD risk than traditional lipids such as HDL-C, LDL-C, and TG. A meta-analysis suggests that walking can significantly lower non-HDL-C levels.⁷ However, there are very few studies (with unequivocal results) that have evaluated the effect of aerobic exercise on non-HDL-C levels¹ and more studies are required to delineate the same.

3. Factors influencing effect of exercise on lipids

Factors such as training time or training intensities can have different effects on lipid levels. Studies suggest that exercise time, and exercise intensity and energy consumption all have an effect on exercise-induced changes in blood lipids.¹ Generally a mix of aerobic and resistance training is recommended and a gradual build-up of intensity and duration is advisable, especially in patients with CVD who may have limited exercise capacity and/or other risk factors. HDL-C is more sensitive to exercise than LDL-C and TG. Patients already taking statins may require additional oversight for muscle pain, etc.

Exercise has beneficial roles in autonomic function, can improve insulin sensitivity, normalize elevated blood pressure, promote endothelial nitric oxide production and improve leptin sensitivity

4. Role of yoga

Several controlled trials have demonstrated modest improvement in lipid profiles by the practice of yoga for 6 weeks to 2 years in healthy subjects and in patients with CVD. Yoga has traditionally been equated with mild exercise. According to the most recent physical activity research, the metabolic equivalents (METs) of Hatha yoga have been estimated to be 2.5 METs.⁸ When the sun salutation (Surya Namaskar) is added, the METs increased to 3.74 which is equivalent to moderate intensity exercise.⁹ Although there are no large studies available, the current guidelines encourage yoga practice among Indians due to its numerous cardioprotective effects including possible lipid improvement.

5. General guidelines

Weight loss is known to improve dyslipidaemia and associated CVD along with a range of other co-morbidities. Hence, weight loss, achieved by through a combination of diet, exercise and behavioural strategies is key to tackling CVD as well as promoting health. Current guidelines recommend at least 150 min of moderate-intensity or 75–150 min of vigorous intensity aerobic exercise spread over 3–5 days per week. Additionally, at least thrice a week of resistance training, is recommended. For those who look to maintain weight loss, higher levels of

exercise (225–420 min/week of moderate intensity exercise) is recommended.¹⁰

According to the ESC 2020 guidelines on sports cardiology and exercise in CVD patients, resistance training three times per week, in addition to moderate or vigorous aerobic exercise, should be included in obese individuals with dyslipidaemia to reduce CVD risk.¹¹

Those patients who are not physically active or beginning after a long gap, should initiate their exercise regimes with moderate intensity exercise sessions for short duration. This may then be slowly built up over time as per capacity and comfort of the patient. Patients taking statins for dyslipidaemia may experience muscle pain and soreness, limiting their ability to exercise. In such cases, it is recommended to temporarily discontinue the drug before resuming the challenge or introducing another lipid-lowering agent.

6. Conclusion

Exercise is crucial in the management of CVD and offers significant additional health benefits. In patients with dyslipidaemia, exercise particularly lowers TGs and raises HDL-C while favourably altering more atherogenic small, dense LDL fraction towards larger LDL particles. Patients with CVD risk should undertake at least 150 min of moderate-intensity exercise preferably spread over 3–5 days per week. Additionally, this should be complemented by resistance training 2–3 times per week at a moderate intensity. Slow initiation and gradual build up are important and patients on statins should be observed for development of muscle pain or weakness.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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